

Problem Understanding Document

SC-18: The Retail Relay — Smart Campus Commerce & Resale Ecosystem

Domain: Smart Campus | Date: 6 April 2026 | Venue: SR University

1. Problem Overview

Campus bookstores and convenience stores at SR University face severe operational inefficiencies that directly harm the student experience. The core issue is a disconnect between inventory visibility, purchasing workflows, and the informal economy of used goods that already exists among students.

Students routinely face three compounding problems during semester start and peak periods:

- Waiting up to an hour in queue just to purchase a lab manual or branded item, with no way to check stock beforehand.
- Discovering items are 'Out of Stock' only after arriving at the store, with no notification system to alert them when items return.
- Being forced to buy new copies of textbooks and equipment because there is no verified platform for students to exchange used items.

2. Identified Pain Points

Pain Point	Student Impact	Severity
No real-time inventory visibility	Students travel to store only to find items out of stock	HIGH
No used book / gear exchange system	Students forced to buy new copies every semester	HIGH
Long queues at semester start	1-hour waits to purchase a single item	HIGH
No stock alert / notification system	Students miss restocks and lose access to materials	MEDIUM
No digital textbook management	Physical dependency creates access bottlenecks	MEDIUM
No analytics for store managers	Overbuying and underbuying of stock continues each year	MEDIUM

3. Root Cause Analysis

The problems above stem from a single structural gap: the campus retail system operates as an isolated, offline process with no digital integration. Specifically:

- The inventory database is not exposed to students or staff digitally — all stock queries require physical presence.
- There is no identity-verified marketplace for peer-to-peer transactions, forcing students to rely on informal WhatsApp groups with zero trust or accountability.
- Store managers have no data pipeline for demand forecasting, leading to repeated over/under-stocking cycles each semester.
- The absence of a pre-order or 'Starter Kit' system means every semester begins with a predictable but unmanaged demand spike.

4. Problem Scope & Boundaries

In Scope

- Real-time inventory search for the official bookstore.
- P2P used book and gear marketplace with student identity verification.
- Back-in-stock notification system.
- Click & Collect order management to eliminate queues.
- Basic analytics dashboard for store managers.
- Pre-order system for semester-start Starter Kits.

Out of Scope (for this prototype)

- Full payment gateway integration — placeholder logic only.
- Mobile application — web interface sufficient for demo.
- Physical IoT barcode scanning hardware — logic will be simulated.

5. Stakeholders Affected

- **Primary users — buyers and sellers of both new and used goods.** Students:
- **Need inventory control, demand forecasting, and reduced queue management overhead.** Store Managers:
- **Benefits from reduced waste, circular economy, and student satisfaction metrics.** University Administration:

6. Success Criteria

The proposed solution will be considered successful if it addresses the following measurable outcomes:

- A student can check real-time stock of any item without visiting the store.
- A student can list a used textbook and connect with a verified buyer within the same platform.
- Store managers can view a demand analytics dashboard showing purchase trends by semester.
- Students receive automatic notifications when an out-of-stock item is restocked.
- Queue wait times are reducible via Click & Collect pre-ordering.

